1.

| INSTR | IC | CPi |
|---|---|---|
| FP | $3 \cdot 10^6$ | 1 |
| INT | $1{,}5 \cdot 10^6$ | 2 |
| L/S | $6 \cdot 10^6$ | 2 |
| BR | $1{,}2 \cdot 10^6$ | 3 |

CR = 2 GHz

a) CPU = ? , MIPS = ?

$$CPU = \frac{3 \cdot 10^6 + 3 \cdot 10^6 + 12 \cdot 10^6 + 3{,}6 \cdot 10^6}{2 \cdot 10^9} = \frac{21{,}6 \cdot 10^6}{2 \cdot 10^9} = 10{,}8 \text{ ms}$$

$$MIPS = \frac{IC}{CPU \cdot 10^6} = \frac{\cancel{10^6}(3 + 1{,}5 + 6 + 1{,}2)}{10{,}8 \cdot 10^{-3} \cdot \cancel{10^6}} = \frac{11{,}7}{10{,}8} 10^3 = 1{,}083 \cdot 10^3 = 1083$$

b) $CPi_{INT}$ = ? speedup = 1,5

$$CPU_{new} = \frac{CPU_{old}}{1{,}5} = 7{,}2 \cdot 10^{-3}$$

$$CC_{new} = \frac{CC_{old}}{1{,}5} = \frac{21{,}6 \cdot 10^6}{1{,}5}$$

$$3 \cdot \cancel{10^6} + 1{,}5 \cdot CPi_{INT} \cdot \cancel{10^6} + 12 \cdot \cancel{10^6} + 3{,}6 \cdot \cancel{10^6} = 14{,}4 \cdot \cancel{10^6}$$

$$18{,}6 + 1{,}5\, CPi_{INT} = 14{,}6$$

$\Rightarrow CPi_{INT} < 0$ impossibil

c) CPU ? CPi → INT, L/S ↘ 20 %
→ FP, BR ↗ 10 %

$$CC_{new} = 3 \cdot 10^6 \cdot 1 \cdot 1{,}1 + 1{,}5 \cdot 10^6 \cdot 2 \cdot 0{,}8 + 6 \cdot 10^6 \cdot 2 \cdot 0{,}8 + 1{,}2 \cdot 10^6 \cdot 3 \cdot 1{,}1 =$$

$$= 19{,}26 \cdot 10^6$$

$$\Rightarrow CPU_{new} = \frac{19{,}26 \cdot 10^6}{2 \cdot 10^9} = 9{,}63 \text{ ms} \qquad IMP = \frac{10{,}8}{9{,}63} = 1{,}12$$

| ② | ISA | CCT | $10^{-9}$ | $\frac{1}{CR}$ CR | ALU | L/S | BR | INT | FP |
|---|---|---|---|---|---|---|---|---|---|
| | P1 | 0,5 ms | → | 2 GHz | 3 | 2 | 2 | 2 | 4 |
| | P2 | | | 3 GHz | 1 | 2 | 3 | 2 | 5 |
| | $ic = 2{,}8 \cdot 10^6$ | | | | 20% | 20% | 30% | 20% | 10% |

a) Which is faster (CPU) (MIPS)

$$CPU_1 = \frac{0{,}2 \cdot 2{,}8 \cdot 10^6 \cdot 3 + 0{,}2 \cdot 2{,}8 \cdot 10^6 \cdot 2 + 0{,}3 \cdot 2{,}8 \cdot 10^6 \cdot 2 + 0{,}2 \cdot 2{,}8 \cdot 10^6 \cdot 2 + 0{,}1 \cdot 2{,}8 \cdot 10^6 \cdot 4}{2 \cdot 10^9}$$

$$= \frac{2{,}4 \cdot 2{,}8 \cdot 10^6}{2 \cdot 10^9} = 3{,}36 \text{ ms}$$

$$CPU_2 = \frac{2{,}8 \cdot 10^6 (0{,}2 \cdot 1 + 0{,}2 \cdot 2 + 0{,}3 \cdot 3 + 0{,}2 \cdot 2 + 0{,}1 \cdot 5)}{3 \cdot 10^9} = \frac{2{,}4 \cdot 2{,}8 \cdot 10^6}{3 \cdot 10^9} = 2{,}24 \text{ ms}$$

2 is faster

$$\left.\begin{array}{l} MIPS_1 = \frac{ic}{CPU_1 \cdot 10^6} = 830 \\ MIPS_2 = 1250 \end{array}\right\} \Rightarrow 2 \text{ is faster}$$

b) 1,5 faster CPI ALU

la fel ca sus

c) $\frac{2}{3}$ int $\longrightarrow \frac{2}{3} \cdot 0{,}2 \rightarrow 0{,}13$ 13,33%

25% mai puține instrucțiuni FP → 10 + 6,66 = 16,66%

$$IC = 2{,}8 \cdot 10^6 \cdot 0{,}75 = 2{,}1 \cdot 10^6$$

$$speedup_1 = \frac{CPU}{CPU_{new}} = \frac{3{,}36}{}$$

$$CPU_{new_1} = \frac{2{,}1 \cdot 10^6 (0{,}13 \cdot 3 + 0{,}2 + 0{,}3 + 0{,}2 \cdot 2 + 0{,}17 \cdot 4)}{2 \cdot 10^9} =$$

la fel și pt P2

hände (10)

c) $CPI_{Br}$

$CPI_{L/S}$ $\searrow$ 25%

$$CPU_{new} = \frac{CPU_{Br} + CPU_{L/S}}{Speedup} + (100-$$

$$Speedup = \frac{1}{0,25}$$